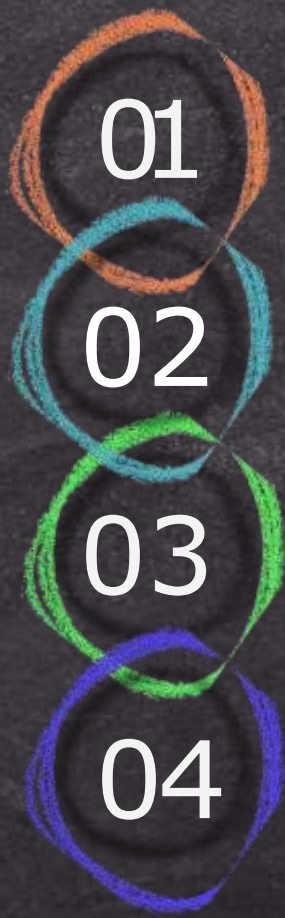
STATISTICAL ANALYSIS FOR INDUSTRIAL ENGINEERING 1

HYPOTHESIS TESTING

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TOPICS



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Null and Alternative Hypothesis

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Null and Alternative Hypothesis

- The null hypothesis is denoted by Ho.
- The alternative hypothesis is denoted by H1.

Example:

Null Hypothesis:

H_0; $\mu = 85\%$

Alternative Hypothesis:

H_1; $\mu \neq 85\%$

Note:

1. Null Hypothesis don't have equalities
2. Two-tailed always have equalities.

Example:

In research,

Ho: There is no significant difference between the academic performance of working and non-working students at EBM University.

H1: There is significant difference between the academic performance of working and non-working students at EBM University.

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Null and Alternative Hypothesis

One-tailed

Two-tailed	Left-tailed	Right-tailed
$H_0: \mu = 23$	$H_0: \mu \geq 23$	$H_0: \mu \leq 23$
$H_1: \mu \neq 23$	$H_1: \mu < 23$	$H_1: \mu > 23$

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Level of Significance

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Level of Significance

- The level of significance is represented by a Greek letter alpha (α**).**
- The commonly used values for α ranges from 0.01 to 0.10 particularly 0.05 or 5% and 0.01 or 1%. The smaller the value of the level of significance the smaller is possibility of rejecting a null hypothesis.
- Level of significance is the rejection region.
- Confidence level is computed as 1- α

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General Procedure of Hypothesis Testing

Step 1: State the Null Hypothesis
Step 2: State the Alternative Hypothesis
Step 3: Set α
Step 4: Collect Data
Step 5: Calculate a test statistic
Step 6: Construct Acceptance / Rejection regions
Step 7: Based on steps 5 and 6, draw a conclusion about H0

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03

Z-Test (One-sample and Two-sample)

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Z-Test (One-sample)

- The simplest and the most common test regarding the significance of sample data is the Z-test. This is used if the sample is equal to or greater than 30.
- The Z-test can be applied in two ways
  - One sample mean test
  - Two sample mean test

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

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Z-Test (One-sample)

Example

A traveler claims that the average weight of his luggage is 40 kilos with an standard deviation of 9.5 kilos. Would you agree to this claim if a random sample of 50 luggage showed an average weight of 38 kilos using 0.05 level of significance.

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Z-Test (One-sample)

Example

A manufacturer claims that the mean weight of their cereal boxes is 500 grams. To test this claim, a random sample of 36 cereal boxes is selected, and their weights are recorded. The sample mean weight is found to be 495 grams with a known population standard deviation of 10 grams. At a 5% level of significance, can you conclude that the mean weight of the cereal boxes is different from the manufacturer's claim?

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Z-Test (One-sample)

Example

A school district claims that the average score of its students in a standardized test is higher than 75. To test this claim, a random sample of 100 students is taken, and their average score is found to be 78 with a known population standard deviation of 8. Conduct a one-tailed z-test at a 5% level of significance to determine if there is enough evidence to support the district's claim.

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Z-Test (Two-sample)

- When two separate samples are drawn at random, these can generate two sample means X1 and X2. Assuming that these samples are taken from normal population, then we can test the difference between the two values X1 and X2 significant or not.

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}$$

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Z-Test (Two-sample)

Example

Fifty full time students were given an examination in Statistics and got an average grade of 95 with a variance of 10.5 while 50 working students have an average grade of 88 with a variance of 8.5. Is there a significant difference in the mean grade of the two groups at 0.05 level of significance?

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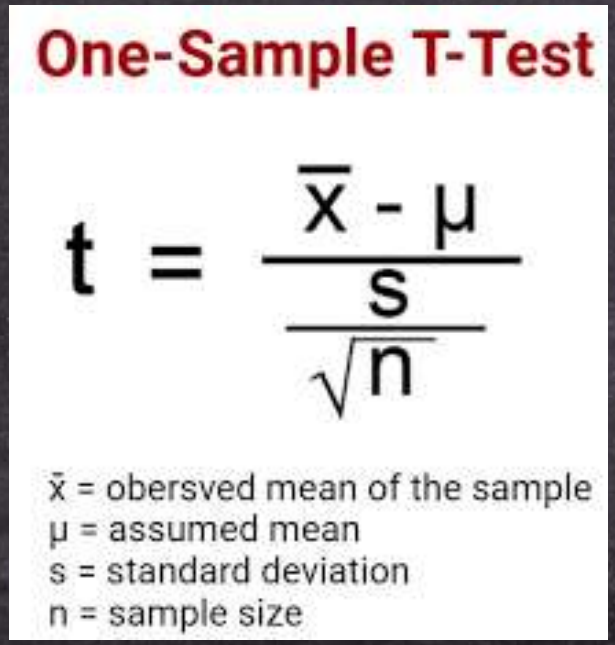


T-test (One and Two Sample Mean Test)

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Z-Test (One-sample)

- To draw inference on a mean in one population where the entries are normally distributed but the variance in not known, a t-test is used.
- This is also used when the sample size is less than 30.



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Z-Test (One-sample)

Example:

A firm produces light bulbs hving an average life span of 725 hours and with a standard deviation of 40 hours. If a random sample of 20 bulbs has an average life span of 700 hours, test the hypothesis that the life span of the lightbulbs is less than 725 hours.

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Z-Test (One-sample)

Example:

A coffee shop claims that the average waiting time for customers' orders is less than 4 minutes. To test this claim, a random sample of 25 orders is taken, and their waiting times are recorded. The sample mean waiting time is found to be 3.6 minutes with a sample standard deviation of 0.8 minutes. Conduct a one-tailed t-test at a 5% level of significance to determine if there is enough evidence to support the coffee shop's claim.

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Z-Test (Two-sample)

- In comparing two sample means, the null hypothesis to be tested are;

Two-Sample T-Test

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{x}_1$ = observed mean of 1st sample
$\bar{x}_2$ = observed mean of 2nd sample
s_1 = standard deviation of 1st sample
s_2 = standard deviation of 2nd sample
n_1 = sample size of 1st sample
n_2 = sample size of 2nd sample

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Z-Test (One-sample)

Example:

Two samples are taken from two groups of students. Group 1 comes from public school and group 2 comes from private school. They were given examination in Chemistry subject and the results are shows in below.

Group 1: n=10, X1bar = 86, s1^2 = 48
Group 2: n=12, X1bar = 88, s1^2 = 38

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Z-Test (One-sample)

Example:

An educational institute is implementing two teaching methods, A and B, for improving student performance in mathematics. They want to determine if method B leads to a higher average test score than method A. A sample of 30 students taught with method A has a mean test score of 75 with a standard deviation of 8. Another sample of 35 students taught with method B has a mean test score of 78 with a standard deviation of 7. Conduct a one-tailed t-test at a 5% level of significance to ascertain if method B results in a significantly higher average test score than method A.

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THANK YOU ☺

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